
Gap Analysis & Feature Improvement Recommendations

GRK Trucking Services Web-based Booking and Tracking Management System
UGRD-ITE6300 Cloud Computing and Internet of Things / UGRD-IT6314 Web Application
Development 1

1. Introduction

The GRK Trucking Services system is a solid B2B web-based booking and tracking platform built with HTML, CSS, JavaScript, PHP, and MySQL, deployed on Amazon EC2 and RDS. It successfully covers the core operational flows: customer booking, administrator fleet/driver assignment, driver job acceptance and status updates, and basic real-time status visibility.

However, when benchmarked against modern logistics platforms (such as Lalamove, which the project explicitly references) and against the stated project objectives of real-time visibility, operational efficiency, and cloud scalability, several practical and competitive gaps remain. This document identifies those gaps and proposes concrete, prioritized feature improvements that can be implemented within the existing technology stack.

2. Highest-Impact Gaps and Recommended Features

2.1 True Real-Time Location Tracking (Biggest Functional Gap)

Current “real-time tracking” is limited to discrete status changes (Pending → Assigned → In Transit → Completed) and the ability to view gate passes or EIR documents. There is no live GPS map, route polyline, or estimated time of arrival (ETA).

Recommended additions:

- Driver location sharing via Geolocation API or background GPS services.
- Live map view (Google Maps, Mapbox, or Leaflet) on Customer Track Shipment and Admin Overview screens.
- Historical route playback and geofencing alerts when the vehicle arrives at pickup or drop-off points.

This directly strengthens the Lalamove-inspired tracking claim and fulfills the project’s real-time status objective more completely.

2.2 Notifications and Communication Layer

No email, SMS, in-app, or push notifications are described for status changes, new job assignments, gate-pass uploads, or delivery delays. Customers and drivers still rely heavily on phone calls for updates.

Recommended additions:

- Automated notifications (email + SMS via Twilio, AWS SNS, or a local SMS gateway) on key booking lifecycle events.
- Simple in-app messaging or chat between Customer, Admin, and Driver.
- Optional WhatsApp Business integration, which is highly relevant in the Philippine market.

2.3 Payment and Invoicing Module

The system currently has no customer-side payment, deposit, invoice generation, or settlement tracking. A payout field exists for drivers, but the revenue collection side remains manual.

Recommended additions:

- Automatic quotation / estimated cost based on vehicle type and approximate distance.

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- Online payment options (GCash, Maya, PayMongo, or Stripe) or a formal “Pay Later / Invoice” workflow for B2B clients.
 - Digital invoice generation and an admin reconciliation dashboard for payments received.

2.4 Mobile Experience and Offline Capability

The system is purely web-based. Drivers operate in the field with variable network connectivity, making a desktop-oriented interface suboptimal.

Recommended additions:

- Progressive Web App (PWA) support with installable homescreen icon, offline caching of active jobs, and background sync.
- Mobile-first responsive redesign of the Driver portal (especially Available Jobs and My Deliveries).
- Optional future native Android companion application for field use.

2.5 Enhanced Proof of Delivery (POD) and Document Handling

Gate-pass viewing and EIR upload already exist, but the documentation and evidence capture capabilities are limited compared with modern logistics platforms.

Recommended additions:

- Photo capture of cargo condition at both pickup and drop-off.
- Digital signature capture on mobile devices.
- Automatic PDF Proof of Delivery generation containing timestamp, location, photos, and signature.
- Per-booking document vault accessible to both customer and administrator.

2.6 Advanced Analytics, Reporting, and Fleet Intelligence

The Admin Overview currently shows only basic counters (total bookings, drivers, fleet status). Deeper operational insight is missing.

Recommended additions:

- Earnings reports, vehicle utilization rates, on-time performance metrics, and popular route analysis.
- Driver performance scorecards.
- Basic predictive suggestions (recommended vehicle type, peak-hour demand indicators).
- Exportable reports (PDF / Excel) and scheduled email digests for management.

2.7 Customer Experience and Trust Features

Recommended additions:

- Post-delivery rating and feedback system (1–5 stars plus optional comments).
- Saved addresses and frequent-route templates for repeat B2B clients.
- Booking templates and support for recurring or scheduled bookings.
- Formal cancellation / reschedule workflow with reason codes.

2.8 Operational and Security Hardening

Recommended additions:

- Two-factor authentication (especially for Admin accounts).
- Comprehensive audit logs recording who changed status, assigned drivers, or edited fleet records.
- Finer-grained role-based access control.
- Basic fleet maintenance tracking (last service date, mileage, next service reminder).
- Soft-delete / archive capability instead of hard deletion of bookings and master records.

3. Prioritization Matrix

The following table ranks the recommended features by business value and estimated implementation effort, assuming continued use of the existing PHP / MySQL / AWS stack.

Priority	Feature	Business Value	Effort
High	Live GPS map + ETA	Core differentiator vs status-only tracking; strengthens Lalamove parity	Medium–High
High	Notifications (Email / SMS)	Reduces phone dependency; improves customer and driver experience	Medium
High	Payment / Quotation module	Enables digital revenue collection and professional invoicing	Medium
Medium	Enhanced POD (photos + signature)	Increases professionalism and reduces delivery disputes	Medium
Medium	PWA / improved mobile driver UX	Critical for field users with intermittent connectivity	Medium
Medium	Ratings + basic analytics	Builds trust and supports continuous operational improvement	Low–Medium
Lower	Fleet maintenance tracking & advanced reports	Long-term operational efficiency and asset management	Medium

4. Conclusion

The current GRK Trucking Services system successfully digitizes the core booking-to-delivery workflow and provides a clean multi-role interface. The gaps identified above are not fundamental flaws; they represent the natural next layer of maturity required to move from a functional academic prototype to a competitive, production-ready logistics platform.

Implementing the high-priority items—live GPS tracking, notifications, and a payment/quotation module—would deliver the greatest immediate improvement in customer experience, operational transparency, and revenue capture, while remaining fully compatible with the existing PHP, MySQL, and AWS technology stack. Subsequent iterations can then address mobile optimization, enhanced proof of delivery, analytics, and security hardening.

These enhancements would close the most noticeable differences between the present system and modern logistics platforms, strengthen alignment with the project’s original objectives, and position GRK Trucking Services for scalable growth within Metro Manila’s expanding logistics market.

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